10 Sovereignmemory

Sovereign Memory Benchmark — ESPHOME

Company: Anticloud FZ LLE

Performance

Operation Time
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remember() (write+encrypt) 8.2ms
recall() (decrypt+lookup) 3.1ms
search_text (1K entries) 4.8ms
search_semantic (1K entries, 384-dim) 12ms
Reload from disk (10K entries) 340ms

Encryption

Property Value
--- ---
Algorithm AES-256-GCM
Key derivation scrypt (N=16384, r=8, p=1)
Nonce 96-bit random per write
Auth tag 128-bit GCM
Password → key time ~200ms (intentionally slow)

Storage

Each entry: ~500 bytes average (compressed JSON + encryption overhead).
10,000 entries ≈ 5MB. Unlimited retention — local disk only.

Data Sovereignty

- Zero cloud sync
- Zero telemetry
- Works fully offline forever
- Password known only to deploying organization